

Darrin Bright

Medical Image Analysis | Computer Vision | Deep Learning

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Research Interests

Medical Image Analysis | 3D Vision | Representation Learning | Weakly Supervised Learning

Education

Vellore Institute of Technology (VIT)

Integrated Master of Technology in Software Engineering, CGPA: 8.81/10

Vellore, India

Sep 2022 - May 2027

Relevant Coursework: Artificial Intelligence, Machine Learning, Statistics for Engineers, Soft Computing, Data Mining Techniques, Applied Discrete Mathematical Structures, Applications of Differential and Difference Equations.

Publications

PortionNet: Distilling 3D Geometric Knowledge for Food Nutrition Estimation

11th Annual Conference on Vision and Intelligent Systems (CVIS), 2025 [Paper] [Code]

A Lightweight Histology-Aware Learning Framework with Spectral Channel Recalibration and Prototype-Graph Refinement for Oral Squamous Cell Carcinoma Detection

IEEE Access, Under Review

Research Experience

Nanyang Technological University

Research Intern | Advisor: Dr. Si Yong Yeo

Singapore

Jul 2026 - Present

- Developing an end-to-end semi-supervised coronary vessel segmentation framework that treats segmentation as bidirectional correction, with complementary branches to recover missed vessels and remove false detections.
- Achieved preliminary Dice improvements of 2.34% on the XCA134 dataset and 1.63% on the XCAD dataset over the state-of-the-art baseline using only 10% labeled data.

Simon Fraser University

Research Intern | Advisor: Dr. Ghassan Hamarneh

Vancouver, Canada

Dec 2025 - Present

- Developed a segmentation-free 3D deep-learning framework for direct detection of mitochondria-ER contact sites in fluorescence microscopy, including STED and SIM imaging.
- Built a physics-informed synthetic-data pipeline and validated predictions against established biological contact patterns in HT1080 and COS-7 cells.

Indian Institute of Technology Kharagpur

Research Intern | Advisor: Dr. Debanjan Das

Kharagpur, India

May 2026 - Jul 2026

- Developed GraphUNet, representing beat-level ECG/PPG morphology as graphs within a U-Net architecture for cuffless blood-pressure estimation, enabling explicit modeling of physiological relationships across heartbeats.
- Evaluated on VitalDB and PulseDB against published baselines, achieving 4.97 mmHg SBP MAE and 2.77 mmHg DBP MAE under subject-disjoint, calibrated evaluation, demonstrating competitive cuffless BP estimation performance.

Indian Institute of Technology Hyderabad

Research Intern | Advisor: Dr. C. Krishna Mohan

Hyderabad, India

Jan 2026 - Jun 2026

- Built a federated continual learning framework integrating a global synthetic-data anchor with V-Information-guided selective re-distillation to preserve prior-task knowledge without sharing patient data or maintaining a conventional replay buffer.
- Evaluated across five MedMNIST modalities, improving average final accuracy by 32.12 percentage points and reducing average forgetting by 20.26 percentage points over FedCIL.

SPHINX-Net: Histology-Aware Framework for Oral Cancer Detection

Advisor: Dr. Bharanidharan N.

- Developed SPHINX-Net, a lightweight hybrid CNN-Transformer framework for patch-level oral squamous cell carcinoma detection, combining histology-aware multi-view contrastive pretraining, spectral channel recalibration, and mini-batch prototype-graph refinement.
- Achieved 96.30% patch-level accuracy with a 1.75M-parameter model at 1,034 FPS.

PortionNet: Distilling 3D Geometric Knowledge for Nutrition Estimation

Advisor: Dr. Kanchan Keisham

- Developed PortionNet, a cross-modal knowledge distillation framework that transfers 3D geometric knowledge from point clouds to an RGB-only model through a lightweight RGB-to-Geometry Adapter, eliminating depth-sensor requirements at inference.
- Achieved SOTA on MetaFood3D, with 17.43% volume MAPE and 15.36% energy MAPE, reducing errors by 57.9% and 77.4% relative to the previous best; further achieved 12.17% energy MAPE on SimpleFood45 dataset.

Indian Institute of Technology Jodhpur

Jodhpur, India

Research Intern | Advisor: Dr. Deepak Mishra

May 2025 - Jul 2025

- Developed a hierarchical vector quantization framework for skin lesion segmentation, pretraining on unlabeled ISIC 2018 dermoscopy images and using the learned discrete codes to regularize PH2 fine-tuning: 79.76% Dice, a consistent +1.38% over the VQ-free baseline across repeated runs.
- Ran a 13-configuration ablation of vector quantization inside Swin Transformer encoders on Pascal VOC 2012, achieving 74.61% Dice, +7.34% over the CNN baseline and +3.91% over the unquantized Swin.

Technical Skills

Programming Tools

Python | SQL | Git | LaTeX

ML & DL Tools

PyTorch | TensorFlow | OpenCV | Matplotlib | NumPy | Keras | MONAI

Medical Imaging

X-ray | CT | Histopathology | Fluorescence Microscopy | Dermoscopy

Work Experience

Astute

Vellore, India

Project Manager and Technical Lead, Intern

Oct 2024 - Apr 2025

- Architected SEO AgentFlow, a multi-agent AI framework with six specialized agents for keyword extraction, on/off-page optimization, technical SEO, and website analysis.
- Led development of AI-driven automation workflows integrating RAG-based content generation, web scraping, and automated content and advertisement generation.

GenAI/ML Engineer, Intern

Jul 2024 - Oct 2024

- Built a data-driven keyword research pipeline integrating SERP API, Google Trends, and DataForSEO with multi-dimensional ranking for competitive analysis.
- Developed automated advertisement generation using Stable Diffusion for poster design and Gemini for tagline generation and placement.

Leadership and Service

Chairperson, IEEE Robotics and Automation Society, VIT

Dec 2024 - Dec 2025

Volunteer, National Service Scheme

Dec 2023 - Dec 2024

Volunteer, Becoming I Foundation

Dec 2024 - Dec 2025

Honors and Awards

Winner, Project 2039 Hackathon, Vellore Institute of Technology, 2024

Winner, Alphaforge Ideathon, Vellore Institute of Technology, 2024

Runner-up, Startup Demo Day, Vellore Institute of Technology, 2025

Runner-up, Biomimicry Innovation Challenge, Vellore Institute of Technology, 2024

Second Runner-up, HackWar Hackathon, Vellore Institute of Technology, 2025